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Cybercrime awareness and digital self-protection among Austrian secondary school students

Marina Unterweger^{1*} , Corinna Hörmann^{2†}, Lisa Kuka^{1†} and Barbara Sabitzer^{1†}

[†]Corinna Hörmann, Lisa Kuka and Barbara Sabitzer have contributed equally to this work.

*Correspondence:

Marina Unterweger
marina.unterweger@jku.at

¹Department of STEM Education,
Johannes Kepler University Linz,
Altenberger Straße 68, 4040 Linz,
Austria

²Department of Computer Science
Education, University Salzburg,
Jakob-Haringer-Straße 2,
5020 Salzburg, Austria

Abstract

This study investigates Austrian secondary school students' knowledge, experiences, and attitudes toward cybercrime and digital self-protection. Based on a nationwide survey of 656 students, the research offers insights into students' awareness of internet-related threats, including fake news, deepfakes, and other forms of online manipulation. While a large majority reported familiarity with the concept of cybercrime, male students more frequently indicated higher awareness, greater self-assessed preparedness, and more confidence in their ability to protect themselves, whereas female students more often reported uncertainty or limited preparedness. Age-related patterns were weaker but suggested that older students feel somewhat more prepared, while younger peers show greater interest in additional education. Despite these gaps, students expressed a strong interest in receiving more school-based education on online safety and responsible digital behavior. The findings reveal not only a diversity of protective practices among students, but also a strong call for more engaging and practical approaches to digital education. Many students advocated for workshops, expert lectures, and the integration of cybersecurity content into regular subjects. Through descriptive and comparative analysis of the survey data, this study highlights the pressing need to strengthen digital literacy curricula in ways that address gender-specific patterns, acknowledge the perspectives of diverse students, and equip young people with the competencies necessary to navigate the challenges of an increasingly connected world.

Keywords Cybercrime, Digital education, Digital literacy, Cybercrime education, Digital resilience, Artificial intelligence

1 Introduction

The rapid digitalization of modern society has deeply transformed how young people communicate, learn, and interact with others. Online platforms, mobile apps, and digital technologies are now central to the daily routines of adolescents. However, alongside the many opportunities that digital media provide, there are also significant risks. In recent years, cybercrime has become one of the most serious security issues worldwide, posing significant challenges to individuals, organizations, and societies [1]. Adolescents,



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who often lack advanced digital literacy and risk awareness, are particularly vulnerable to these online threats. This vulnerability not only calls for effective prevention but also raises questions about the efficiency of existing educational strategies and the responsibility of schools and policymakers to ensure that interventions are both equitable and impactful.

Cybercrime includes a wide range of offenses that affect young people in different ways. These include psychological and social attacks such as cyberbullying, doxing, and fraping, which can severely harm adolescents' emotional health and social relationships [2–4]. Sexually motivated offenses like cyber grooming, sexting, and sextortion also pose serious risks, often involving manipulation, threats, and blackmail [5–7]. Additionally, fraud-related crimes, such as phishing, identity theft, and romance scams, target young people's limited experience with online deception [8, 9].

Recent statistics from Austria underline the growing threat posed by cybercrime. In 2023, more than 65,000 cases were reported, nearly doubling since 2019 [10]. Across Europe, similar trends are visible. A large cross-national study by the World Health Organization (WHO) highlights rising cyberbullying rates in many European countries [11]. National studies from Switzerland, Germany, and Austria further confirm high levels of cyberbullying and online harassment among adolescents, with many incidents occurring on popular platforms such as Instagram, TikTok, WhatsApp, and Snapchat [12–15].

At the same time, the rapid advancement of artificial intelligence has further complicated the digital landscape, as many adolescents use AI tools such as ChatGPT regularly [12]. Technologies such as deepfakes and automated scams have introduced new risks, making it more difficult for adolescents to distinguish between authentic and manipulated content. This evolving context raises important questions about how well adolescents understand online threats, what protective measures they take, and what role schools can play in fostering digital resilience and responsible online behavior.

To address these issues, the present study surveyed Austrian secondary school students in early 2025. A total of 656 completed responses were collected through an anonymous online questionnaire. The survey explored students' knowledge about cybercrime, their personal experiences with digital risks, their self-protection practices, and their views on how schools can better prepare them for the challenges of the digital world. In doing so, it also examined potential differences by gender and age, and collected qualitative insights into students' expectations for school-based education.

This paper is structured as follows: The next section provides the theoretical framework, defining key forms of cybercrime relevant to young people and presenting recent statistics from Austria and Europe. It also outlines the legal context in Austria and discusses the specific role of schools in both exposing students to digital risks and serving as key sites for prevention and education. Following this, the methods section describes the research design, data collection, and sample characteristics. The results section presents the key findings of the survey, highlighting students' awareness, experiences, protective behaviors, and perspectives on school-based education. The paper concludes with a discussion of the findings, their implications for educational practice, and potential directions for future research. By offering a comprehensive analysis of students' perspectives, this study aims to support the development of more effective, relevant, and student-centered approaches to cybercrime education in Austrian schools.

2 Theoretical framework

2.1 Defining cybercrime

Cybercrime, also referred to as internet crime, encompasses a wide array of criminal activities that are either carried out through, or directed against information and communication technologies (ICT). Despite its widespread usage over the years, the term *cybercrime* remains without a universally accepted definition [1, 16, 17].

The European Commission defines cybercrime as offenses committed through electronic communication networks. It distinguishes among cyber-specific offenses, online fraud, and illegal online content. In response to these challenges, the European Union implements targeted legal frameworks, fosters cooperation among law enforcement agencies and judicial authorities, and promotes partnerships with industry stakeholders. Moreover, the EU advances prevention measures aimed at enhancing the protection of its citizens [18].

At the national level, the Austrian Federal Criminal Police Office (German: Bundeskriminalamt, BKA) categorizes cybercrime into two broad groups: *cybercrime in the narrower sense* and *cybercrime in the broader sense*. Cybercrime in the narrower sense includes offenses that directly target data or computer systems, such as hacking, data destruction, or distributed denial of service (DDoS) attacks. In contrast, cybercrime in the broader sense refers to the use of information and communications technology (ICT) to facilitate traditional crimes, including fraud, drug trafficking on the dark web, online child exploitation, cyber grooming, and cyberbullying [19].

This study focuses specifically on forms of cybercrime that predominantly affect adolescents. Given their intensive use of digital media, young people are particularly vulnerable to certain risks in online environments. Accordingly, this study examines various aspects of cybercrime by classifying them into three main categories:

1. Psychological and Social Attacks: including cyberbullying, doxing, and frapping;
2. Sexually Motivated Cybercrime: such as cyber grooming, sexting, and sextortion;
3. Fraud and Data Theft: including phishing, identity theft, and romance scams.

A recurring theme across all three categories is *cyberbullying*. This phenomenon is frequently intertwined with other forms of cybercrime, as perpetrators may employ tactics such as frapping, doxing, or identity theft to intensify the harassment of their victims. Similarly, sexually motivated offenses, such as sextortion or the unauthorized dissemination of sexting content, can cause profound psychological harm when combined with bullying behavior.

In addition to these interconnections, the evolving role of artificial intelligence (AI) has significantly reshaped the cybercrime landscape. Automated insults, deepfakes, hate speech, and personalized attacks orchestrated through algorithms represent emerging threats that amplify the scope and intensity of cybercrime.

What unites all of these forms is their deliberate targeting of victims, whether through exposure, manipulation, or extortion. Consequently, cyberbullying often emerges as an overarching concept encompassing various forms of digital aggression, thus highlighting the destructive impact of cybercrime on young individuals.

2.2 Youth-relevant cybercrime: forms and students' perceptions

2.2.1 Psychological and social attacks

Psychological and social attacks in digital spaces include various forms of online aggression that many adolescents frequently encounter. *Cyberbullying*, *doxing*, and *fraping* are some of the most common examples, each posing serious risks to young people's emotional well-being and social relationships.

Cyberbullying involves the intentional and repeated use of digital devices such as computers, smartphones, or other electronic technologies to cause harm or distress to others. Although the term has been in use since 1999, there is still no universally agreed-upon definition [20–22]. It is, however, widely understood as a form of repeated harassment carried out through digital communication channels. While cyberbullying shares many similarities with traditional bullying, it is often examined as a separate issue due to its distinct features, such as the anonymity provided by digital platforms and the potentially more severe emotional consequences for victims. Moreover, cyberbullying often overlaps with school-related situations, where offline conflicts may escalate into online harassment, and online attacks can, in turn, affect the daily school life and well-being of young people [2].

Another related phenomenon is *doxing*, which refers to the malicious practice of publishing private or sensitive personal information on the internet without consent. This may involve details such as addresses, phone numbers, personal photos, or even workplace information. The term *doxing* originated in the hacker communities of the 1990s and comes from the phrase “dropping documents”, originally describing actions intended to reveal someone's identity online. Today, doxing is often used to intimidate, blackmail, harass, or publicly shame individuals. A key aspect of this practice is that the intent to harm remains central, regardless of whether the information was already available in isolated sources elsewhere on the internet. The deliberate collection, organization, and disclosure of this information for the purpose of causing damage is what characterizes doxing [3]. In many cases, doxing and cyberbullying are closely connected, as cyberbullies may use doxing to expose private details of their victims, thereby amplifying the harassment or encouraging others to participate in the attacks. *Fraping*, another related form of digital aggression, refers to the unauthorized use of someone's social media account to post messages or content without their consent. Typically, this involves changing the victim's profile or posting misleading, embarrassing, or inappropriate messages under their name. This practice is most common among younger people, particularly within tight peer groups where it is often seen as a joke or prank. While some posts, especially those that are explicitly offensive, are usually removed, others considered amusing or light-hearted may be left visible. However, in more diverse or older social groups, fraping is generally viewed as unacceptable behavior [4]. Fraping demonstrates how the boundaries between humor and harassment can become blurred, particularly in digital settings, and how seemingly harmless pranks can sometimes contribute to broader patterns of online harassment or social exclusion. This ambiguity between playful interaction and harmful behavior is also reflected in broader trends of cyberbullying: according to an Austrian study on cyberbullying among adolescents, 44% stated that the reason for cyberbullying is the inability to recognize the boundary between fun and seriousness [14].

2.2.2 Sexually motivated cybercrime

Sexually motivated cybercrime includes various online risks that adolescents increasingly face in digital environments. Common examples are *cyber grooming*, *sexting*, and *sextortion*, all involving the misuse of sexual content or intentions to manipulate, pressure, or exploit young people, often leading to serious emotional harm.

Cyber grooming refers to the deliberate process by which adults use the internet to establish contact with minors for sexual exploitation. Originating from the English term “to groom”, this practice involves careful manipulation, where offenders often hide behind false identities to build trust with children and adolescents. Their ultimate goal can range from persuading minors to engage in sexual acts via webcam to arranging in-person encounters. Adolescents between the ages of 10 and 15 are particularly vulnerable to such approaches, as they are in a sensitive stage of development and may struggle to recognize the risks [5]. Offenders typically target platforms popular among young people, such as social media networks and chat rooms, which offer easy access and anonymity [23]. While cyber grooming and cyberbullying are distinct, they share key similarities, including the deliberate use of digital tools, power imbalances between perpetrators and victims, and repeated patterns of harmful behavior. Victims of cyberbullying may be especially susceptible to grooming, as offenders sometimes exploit materials shared during bullying episodes to manipulate or blackmail their targets. Studies show that socially isolated adolescents or those facing difficulties with peers are at greater risk of both cyberbullying and grooming, as offenders often pose as empathetic friends to gain trust and exert control [23].

Sexting refers to the voluntary exchange of sexually explicit or suggestive messages and images via smartphones or digital platforms. The term, which combines “sex” and “texting”, first appeared in the media in 2005. While sexting itself is not necessarily classified as cybercrime, it can easily lead to criminal situations, particularly when private content is shared without consent or used for harassment, blackmail, or public humiliation [6]. Public discourse often portrays sexting as a risky behavior among adolescents, as many fail to fully understand the possible consequences of sharing intimate content.

Closely connected to these issues is *sextortion*, a serious form of digital extortion that involves threatening to release sexual images or videos to force victims into specific actions. The term, combining “sexual” and “extortion” [24], describes cases where perpetrators use intimate material to pressure victims into providing additional explicit content, engaging in sexual acts, or meeting other demands. Unlike sexting, which involves voluntary participation, or revenge porn, which focuses on unauthorized distribution out of revenge, sextortion centers on the use of threats to control the victim—even if the images are never actually published [7]. This form of coercion places victims under extreme psychological pressure and is increasingly recognized as one of the most serious online threats faced by adolescents.

2.2.3 Fraud and data theft

Among the forms of cybercrime affecting adolescents, fraud and data theft play a significant role. These offenses often exploit adolescents’ limited digital literacy and inexperience. Driven by curiosity and social interaction, young people are particularly vulnerable to tactics such as phishing, identity theft, and romance scams.

Phishing is a common method where attackers use fake emails, websites, or messages to steal sensitive information such as passwords or credit card data. The term “phishing”, derived from “fishing”, reflects the act of luring victims [8]. Among adolescents, phishing is not limited to financial fraud but can also serve as a bullying tactic. Through deception and manipulation, offenders convince young people to reveal personal or financial information about themselves or their families. This data is then misused to access social media accounts, abuse passwords, or make unauthorized purchases with the victim’s or their parents’ credit cards [25]. Such incidents can cause emotional harm and social repercussions, making phishing prevention an important topic for schools.

Closely related to phishing is *identity theft*, or “impersonation”, where offenders assume someone else’s identity to cause harm or gain benefits. This may involve sending or posting content under another person’s name, damaging their reputation, endangering them, or harming relationships [26].

Another manipulative method is the *romance scam*, where offenders create fake profiles on social media or dating platforms to build emotional connections with victims. Over time, they gain the victim’s trust to solicit money or gifts. This scam is particularly harmful due to the dual trauma it inflicts—financial loss and emotional betrayal—and the shame that often prevents victims from reporting such cases, leading to underreporting [9]. While romance scams mainly target financial gain, they can also involve cyberbullying when offenders use romantic or friendly connections to collect sensitive information, later using it for blackmail or humiliation.

These examples highlight the need for preventive education in schools, focusing not only on technical security measures but also on raising awareness about psychological manipulation and fostering critical digital literacy.

2.2.4 Students’ perceptions of cybercrime and online risk

Building on the overview of youth-relevant cybercrime in the preceding subsections, it is important to note that exposure to online risks alone does not determine outcomes. Empirical research shows considerable variation in how adolescents experience online risks, how they respond to them, and the extent to which they adopt protective behavior.

Large-scale European evidence from the EU Kids Online 2020 survey illustrates this variation clearly. Across participating countries, between 7 and 45% of children aged 9–16 reported that something online had bothered or upset them during the previous year, although substantially higher proportions reported exposure to potentially risky activities such as sexting or communicating with unknown persons online. Among victims of online bullying, some reported no harm while others experienced strong distress, indicating differing subjective experiences of similar situations. Children most often disclosed negative experiences to parents or friends rather than teachers or professionals, and many relied on coping strategies such as blocking users, closing applications, or ignoring incidents. Most respondents nevertheless stated that they generally knew how to react to unwanted online behavior, suggesting considerable diversity in coping responses among young users [27].

European cybersecurity education research further highlights differences between adolescents’ perceived confidence and their actual cybersecurity practices. Findings from the CONCORDIA “Teaching Cybersecurity in High-School” survey in 2021 indicate that while many high-school students report feeling confident in recognizing and

handling online risks, a substantial proportion fail to select recommended security practices when confronted with concrete cybersecurity scenarios. Teachers and parents frequently expressed greater concern regarding students' preparedness than students themselves, suggesting that perceived competence may not fully reflect effective risk management skills. The report therefore emphasizes the need for structured cybersecurity education and stronger collaboration between students, educators, and families in addressing online risks [28].

Similar patterns emerge in studies examining digital skills and online engagement. Using survey data from teenagers in the United Kingdom, Livingstone and Helsper found that online access, use, and skills mediated relationships between demographic characteristics and experiences of both online opportunities and risks. Their analysis further identified a positive relationship between opportunities and risks, indicating that increased participation in digital environments may simultaneously expand beneficial experiences and exposure to potentially harmful content [29].

Evidence from international educational assessments likewise highlights differences in students' digital competence. According to the International Computer and Information Literacy Study 2023 (ICILS), approximately 43% of students in the European Union demonstrate only limited levels of computer and information literacy. The results also indicate substantial disparities between countries and within national education systems, as well as strong associations with socioeconomic background and home learning resources [30]. These findings suggest that students' ability to recognize and manage online risks may vary considerably depending on digital competence and educational context.

Survey-based awareness studies conducted in different national settings report heterogeneous levels of cybercrime knowledge and protective practices. A mixed-methods study among youths in Kaduna State, Nigeria, found that 83% of respondents reported awareness of various cyber threats, including cyberbullying, identity theft, hacking, and phishing, and participants described preventive practices such as software updates, password management, and monitoring online accounts [31]. Similarly, a youth awareness survey in Delhi NCR reported that 98% of respondents indicated familiarity with cybercrime or cyberterrorism [32]. In contrast, research involving 600 students found that many respondents rarely changed passwords, showed limited awareness of two-factor authentication, and paid little attention to browser security or verification of website authenticity, indicating gaps between general awareness and specific protective practices [33]. Taken together, these studies indicate that while general awareness of cybercrime terminology is often high, detailed knowledge of protective practices and risk recognition varies considerably across educational and cultural contexts.

Research conducted among senior high school students further demonstrates differentiated awareness across cybercrime types. While respondents reported high awareness of cyberbullying, cyberpornography, and hacking, understanding of identity theft and online scams was comparatively lower [34]. Qualitative interviews with young cybercrime victims likewise found that participants frequently reported experiencing cyberbullying, hacking, and blackmailing, and some indicated that online decisions were made without prior awareness of potential cybercrime risks [35]. Studies among university students additionally report high levels of cybercrime risk awareness alongside only weak relationships between awareness and attitudes toward internet use [36].

Protection Motivation Theory (PMT) provides a framework for understanding how such differences in awareness and response may influence protective behavior [37]. This framework distinguishes between threat appraisal and coping appraisal processes preceding action. Threat appraisal involves perceived severity and susceptibility, as well as perceived rewards associated with maintaining risky behavior. Coping appraisal considers self-efficacy, response efficacy, and response costs. Applied to cybercrime contexts, protective actions such as reporting incidents, managing privacy settings, or preserving evidence are more likely when threats are perceived as personally relevant and safeguards are considered effective and manageable.

Technology Threat Avoidance Theory (TTAT) extends these appraisal processes to digital environments by emphasizing situational and technological influences on avoidance decisions [38]. Factors such as platform familiarity, interface design, message complexity, and convenience-security trade-offs (situations in which users prioritize ease of use or speed over stronger protective measures) may shape how users evaluate warnings or potential threats. For example, students may reuse passwords, quickly accept permissions, or ignore security warnings because safer alternatives require additional effort or interrupt ongoing activities. TTAT therefore offers a complementary perspective for understanding how students' judgments about safeguards and usability influence cybersecurity behavior in everyday digital environments.

Taken together, empirical studies demonstrate substantial variation in adolescents' cybercrime awareness, experiences, and coping responses across contexts. PMT and TTAT provide complementary theoretical perspectives for examining how these differences in perception and technological context may shape protective behavior in online environments.

2.3 Cybercrime statistics: Austria and Europe

Cybercrime has become one of the biggest security problems in Austria and Europe in recent years. As people use the internet more and more, digital crime has also increased. According to the *Cybercrime Report 2023* from Austria's Federal Criminal Police Office (German: Bundeskriminalamt), there were 65,864 reported cases of cybercrime in Austria in 2023. This is an increase of 9.4% compared to the previous year. Since 2019, the number of reported cases has almost doubled. This sharp increase illustrates the growing severity of cybercrime in Austria. The main reason for this increase is the ongoing digitalization of many areas of life, which creates new chances for criminals to commit online crimes. Even though the number of cases has gone up, police were able to solve 440 more cases compared to the year before. This shows that some progress has been made in fighting cybercrime. However, the high number of cases remains a big challenge, not only for police, but also for society as a whole. To better fight cybercrime, Austria has started a major reform of its criminal services. This reform will create 700 new jobs, with 300 of them focused on cybercrime and organized crime. In addition, the *Cyber Crime Competence Center (C4)* will become its own department inside the Federal Criminal Police Office. This change will help police better respond to the growing digital crime threats. The main goals of this reform are better prevention, stronger international cooperation, and improved training for investigators [10].

In Europe, cybercrime and cyberbullying are also increasing problems. The World Health Organization (WHO) Regional Office for Europe published a large study in 2024

called HBSC (Health Behavior in School-aged Children). It looked at bullying, cyberbullying, and violence among young people in 44 countries and regions. The study found that while traditional bullying in schools has stayed about the same, cyberbullying has gone up. About 12% of young people said they had bullied others online, and 15% said they had been victims themselves. Both boys and girls were affected in similar numbers: 15% of boys and 16% of girls reported being victims of cyberbullying. These numbers have increased since 2018. The study also showed that boys are more likely to use physical violence, while girls are more often involved in cyberbullying. The WHO recommends that governments, schools, and families work together to improve online safety and teach young people about empathy and respectful behavior online [11].

In Switzerland, the *AMES Study 2024* also showed how common cyberbullying has become. The study asked more than 1000 young people aged 12–19 from all language regions of Switzerland about their experiences. About 24% said they had been insulted or harassed in private messages in the past two years. Public cyberbullying was less common, with 7% saying they were bullied or insulted in public online spaces. Other common experiences included being excluded from online groups (20%) and having embarrassing photos shared online (20%). Around 7% said they were pressured or threatened online. Boys were more often involved in public cyberbullying than girls. The study also found that 43% of young people indicated to insulting others in private messages, and 20% said they had made fun of others online. Boys were more likely to be involved in these actions. Another interesting result from the study is the strong use of artificial intelligence (AI) tools. Around 71% of young people said they had already used AI tools such as ChatGPT. About one-third of them said they use these tools at least once a week. This shows the importance of teaching young people to check online information carefully and to think critically about digital content [12].

Germany also faces serious challenges with cyberbullying. The *Cyberlife V study* from 2024 showed that 18.5% of students said they were victims of cyberbullying. This means more than two million young people in Germany have been affected. The rate has been rising steadily over recent years. The study also found mixed results regarding prevention in schools. About 55% of students said their schools teach them how to deal with cyberbullying, but only 28% felt supported by their schools when they experienced it. This shows that many schools still struggle to deal with the problem effectively [13].

In Austria, cyberbullying is also a serious issue. A 2022 study by Saferinternet.at and ISPA looked at cyberbullying during the COVID-19 pandemic. The study asked 400 young people aged 11–17 about their experiences. About 17% said they were victims of cyberbullying, and 42% said they had seen others being bullied online. Common types of bullying included insults (48%), ghosting (46%), and spreading false information (41%). Most incidents happened on Instagram and TikTok [14]. The *Youth Internet Monitor 2025* showed that these platforms, along with WhatsApp and Snapchat, are used daily by most young people in Austria [15].

The study also found that many young people bully others because they want power or because they do not realize when “fun” turns into serious bullying. Most victims knew who their bullies were, which means that many cases are not anonymous. Many young people said that adults were often not helpful in these situations. Nearly half said that adults did not take the problem seriously, and one-third said teachers ignored the issue [14]. This shows the need for better prevention, more training for teachers, and stronger

support systems in schools. Also, social media platforms need to improve their reporting systems to help young victims more effectively.

Altogether, the statistics from Austria and across Europe show that both cybercrime and cyberbullying are growing problems that require ongoing efforts from governments, schools, families, and technology companies.

2.4 Legal context in Austria

Austria has strengthened its legal framework in recent years to better address cybercrime and protect its citizens. A key law is the *Cyberbullying Law* (§ 107c StGB), introduced in 2016. It specifically targets ongoing harassment through digital channels, such as social media or messaging services, and allows for criminal prosecution of such acts. In addition to this law, the Cybercrime Competence Center (C4) serves as Austria's central unit for cybercrime investigations and prevention. Citizens can also report online crime through the official reporting office. However, formal complaints are still required to be filed in person at a police station [19].

Besides the Cyberbullying Law, several other criminal laws in Austria can apply in cases of cybercrime. These include regulations concerning defamation, insults, false accusations, child protection, illegal image sharing, and data misuse. For example, laws against defamation and insults may apply to cases of online slander or threats, while child protection laws address offenses such as possession or distribution of illegal content involving minors. Other laws cover illegal recording of images and the misuse of personal data in digital environments. Together, these legal provisions allow Austrian authorities to respond to various forms of cybercrime, from online harassment to data-related offenses. However, as digital technologies continue to develop, ongoing legal updates and preventive efforts remain essential [39].

2.5 Cybercrime in the school context

Schools represent critical environments where students' increasing engagement with digital technologies creates both educational opportunities and significant vulnerabilities to cybercrime. As digital tools become embedded in students' academic and personal lives, they access online spaces for social interaction, creativity, and learning. While these practices foster autonomy and innovation, they also expose learners to various risks, including malware, inappropriate content, and online harassment [40]. According to OECD [40] (Organisation for Economic Co-operation and Development), a notable proportion of young individuals report negative online experiences, particularly cyberbullying. This includes non-consensual sharing of photos or videos, social exclusion from digital groups, verbal abuse, and online threats. Moreover, the proliferation of advanced technologies such as Artificial Intelligence (AI) and the Internet of Things (IoT) has amplified these risks. Profiling, location tracking, and unauthorized recording through smart devices raise profound concerns regarding privacy and safety. While research into the effects of digital environments on children's mental health is still emerging, concerns about their well-being in online spaces are increasingly shared by the public [40]. In parallel, educational institutions themselves are now targets of cyberattacks, given their role in storing sensitive data about students and families [41].

In response to these challenges, schools bear a dual responsibility: safeguarding students and providing them with the necessary knowledge and competencies to navigate

digital environments securely. Educational policy and digital service design must take into account children's cognitive and developmental capacities, particularly regarding their digital literacy and ability to provide informed consent [40]. Schools serve as key institutions for promoting digital safety and (digital) resilience through both preventive and responsive strategies. These include educating students about online behavior, establishing clear policies, and supporting affected individuals. Structural measures such as institutional acceptable use policies and comprehensive digital citizenship programs are essential to ensure coordinated defenses against cyber threats [42]. National policy frameworks are likewise expected to address core principles including privacy, access, non-discrimination, and appropriate technology use [40].

Globally, frameworks such as the European Commission's DigComp provide a comprehensive structure for developing essential digital competences among students and educators. DigComp defines five core areas—information and data literacy, communication and collaboration, digital content creation, safety, and problem-solving—emphasizing not only technical skills but also critical thinking, ethical awareness, and responsible digital engagement [43]. These competences are increasingly vital in AI-driven societies, where skills such as computational thinking, big data literacy, and critical evaluation of digital content are central to active civic participation and future employability [42, 44].

The Austrian curriculum "Digitale Grundbildung" (Digital Basic Education) builds directly upon the national version of the DigComp framework (DigComp 2.2 AT) [45] and exemplifies how such European-level guidelines are operationalized within national education systems. As a compulsory subject in both Compulsory Secondary Schools (German: Mittelschulen) and Academic Secondary Schools (German: Allgemeinbildende Höhere Schulen), the curriculum aims to equip students with the skills to safely and critically engage with media and digital technologies. It includes explicit reference to student protection and digital rights. Key learning outcomes include competencies in data handling, information systems, cybersecurity, and the ethical use of digital media. Students explore topics such as data misuse, online surveillance, malware protection, and common cyber threats including cyberbullying, grooming, and identity theft. Didactically, the curriculum emphasizes reflective and critical engagement with digital tools, fostering students' ability to responsibly shape and question digital environments [46].

In practice, teachers, school counsellors, and external partners play a pivotal role in translating educational frameworks and curricula into meaningful classroom implementation. They are central to a whole-school approach to cyber safety and digital literacy. Beyond delivering curriculum content, teachers also guide ethical decision-making and serve as key communicators in raising families' awareness of digital risks [47]. Professional development is essential to ensure educators remain equipped to teach emerging topics such as AI-generated misinformation, plagiarism detection, and privacy management. Head teachers and school leadership influence the successful implementation of technology by setting supportive conditions and fostering a culture of digital responsibility [42]. Partnerships between educational institutions, civil society, and technology providers contribute to broader system-level responses by offering context-specific support and strategic resources [41].

In this context, the concept of digital resilience has gained prominence. Defined as the capacity to manage and recover from digital risks while maintaining well-being, digital resilience involves more than just protective strategies; it promotes critical thinking,

emotional regulation, and adaptive coping [48]. As resilience is often acquired experientially rather than through direct instruction, educational environments should create safe opportunities for students to reflect on and learn from negative online experiences [49]. Educational systems thus play an essential role in cultivating resilience through the promotion of social-emotional learning, digital citizenship, and ethical awareness. Programs like “Digitale Grundbildung” explicitly support these goals by encouraging critical media reflection and the recognition of normative structures in digital environments [46]. These competencies are further reinforced through supportive peer structures and open communication channels with educators [49].

A coordinated, whole-school approach—involving educators, families, policymakers, and external experts—is necessary to foster a resilient and critically engaged student body. Schools must not only teach digital skills but also promote a culture of respect, mutual care, and reflective technology use [41, 42]. By embedding these principles into policies, curricula, and everyday school practice, educational institutions can equip students to face the challenges of a digital world responsibly and confidently.

2.6 The role of generative AI in emerging cybercrime risks

The rapid development of Artificial Intelligence (AI) technologies has introduced a new paradigm in various societal sectors, including education and cybersecurity. Central to this transformation is Generative Artificial Intelligence (GenAI), which refers to AI systems designed to create novel content, such as text, images, audio, or video, by learning complex patterns from vast datasets. Unlike earlier AI forms that primarily analyze or categorize existing data, GenAI models, particularly Large Language Models (LLMs), predict sequences to generate original outputs that often closely resemble human-created content [50]. Notable examples include ChatGPT, DALL-E, and Google Bard [50, 51]. These systems, underpinned by deep learning and neural networks, automate the creation of diverse and engaging content, from creative arts to data synthesis [44, 50].

The relevance of GenAI to youth is profound, as adolescents are increasingly exposed to, interact with, and even become creators of AI-generated content. This pervasive digital presence necessitates a critical understanding of both the beneficial applications and the severe risks posed by GenAI, particularly in the context of cybercrime. Young people’s online activities, including sharing innocuous images, can inadvertently expose them to new forms of exploitation [52]. For instance, “nudity” apps, which utilize GenAI to generate non-consensual explicit images from regular photos, are widely accessible and popular among minors, creating significant concerns for online child sexual exploitation and abuse (CSAM—Child Sexual Abuse Material) [52, 53].

Key AI technologies and their cybercriminal applications are fundamentally reshaping the cyber threat landscape, enhancing the speed, scale, and sophistication of criminal activities [50, 54]. Criminal networks are misusing LLMs to craft highly convincing fraud messages, generate “how-to” guides for illicit activities, and automate phishing campaigns [54, 55]. This automation makes cyber-attacks more scalable and efficient [50]. Furthermore, deepfake technology, capable of replicating voices, images, and videos, amplifies threats such as fraud, extortion (sextortion), and identity theft [50, 55]. The proliferation of AI-generated CSAM is particularly alarming. This includes content created by text-to-image models from prompts, the modification of existing CSAM, or the “de-aging” of adult images. Such material can be indistinguishable from real content,

complicating efforts to identify actual victims and leading to re-victimization [53]. The commercialization of non-consensual intimate imagery created by GenAI, with some providers monetizing synthetic content and attracting millions of visitors, highlights a concerning trend where profit motives drive further exploitation [52].

The impacts on adolescents are severe, leading to psychological manipulation, increased vulnerability to exploitation, and significant threats to digital trust. Victims can suffer social anxiety, worry, depression, and a profound erosion of privacy boundaries and trust when their likeness is used to create and distribute fabricated content without their consent [52]. The difficulty in detecting manipulated content exacerbates these psychological harms, as digital artifacts can persist indefinitely, potentially re-victimizing individuals over their lifetime [52, 53]. Criminal networks also increasingly instrumentalize young perpetrators, recruiting them via social media and gaming platforms, involving them in illicit activities, and exploiting their relative impunity [54].

Addressing these challenges requires navigating complex legal and ethical considerations [52, 56]. A key hurdle is the lack of explicit legislation specifically prohibiting the creation and distribution of AI-generated explicit content, leading law enforcement to address cases on an individualized basis using existing, often inadequate, laws [53, 56]. This legislative lag is attributed to the rapid pace of technological advancement, jurisdictional complexities, and concerns about stifling innovation versus ensuring safety, transparency, and ethical use [53, 56]. While some countries are updating definitions to include AI-generated CSAM, a harmonized global approach is lacking [53]. Ethical frameworks emphasizing data privacy, consent, and the responsibility of AI model developers and platform hosts are crucial but remain difficult to enforce, given current legal structures [52].

The implications for school-based education are substantial. While GenAI offers opportunities for personalized learning and content generation [51], it introduces significant challenges related to academic integrity and student well-being [51]. Schools face the dilemma of integrating AI effectively while combating plagiarism and ensuring the development of critical thinking skills [51]. There is an urgent need to enhance digital literacy programs in K-12 schools, teaching students to identify deepfakes and understand the risks of AI-generated content, sexual coercion, and sextortion [56, 57]. Educators require specific digital competences to navigate GenAI for teaching, assessment, and administrative tasks, encompassing skills in digital resources, teaching, learning, assessment, empowering learners, and facilitating learners' digital competence [57]. Schools must develop robust internal guidelines that align with regulations, promote responsible AI usage, and offer professional development for educators [51]. Moreover, increased training and tools are required for both, law enforcement and caregivers, to identify and address AI-generated threats, as schools are often the first to encounter such incidents involving minors [53, 54].

3 Methods

3.1 Research design

The present study employed a quantitative research design using a standardized online questionnaire. This methodological approach was chosen due to its efficiency in reaching a large sample and its high level of acceptance among younger target groups. The questionnaire was developed and administered using the LimeSurvey online platform.

The survey was divided into two main thematic sections:

1. Artificial Intelligence in Schools, and
2. Cybercrime and Internet Safety.

Given that this study focuses on cybercrime in educational contexts, the analysis presented here is based solely on data from the second section of the survey. This part of the questionnaire gathered information regarding students' knowledge, experiences, protective behaviors, and expectations regarding school-based education on internet-related crimes.

In detail, the cybercrime and internet safety section consisted of nine items addressing students' awareness of cybercrime, prior school-based exposure, personal or indirect experiences, familiarity with specific online threats (e.g., fake news or deepfakes), perceived self-protection knowledge, protective online behaviors, and educational expectations. The questionnaire included closed-ended yes/no and multiple-choice items, multiple-response checklists, as well as open-ended items allowing participants to elaborate on their perspectives. Demographic information on age, school type, and gender was collected in a separate section of the questionnaire.

Content validity was supported through the systematic alignment of questionnaire items with the study's research questions and key domains identified in the literature on cybercrime awareness and digital literacy education. The instrument combined structured response formats with open-ended questions to capture both measurable response patterns and contextual student perspectives. Before full implementation, the questionnaire was reviewed internally within the research team's Teaching and Learning Lab, involving both researchers and university student staff members as age-proximate reviewers. Given their recent experience of secondary education and familiarity with the digital environment of the target group, these reviewers were considered suitable for assessing item clarity, wording, and comprehensibility. This process resulted in minor revisions before data collection. Most items assessed distinct constructs with a single question rather than multiple questions combined into a scale (e.g., one item each for self-protection knowledge, preparedness, and school-based education). As standard reliability statistics such as Cronbach's alpha require several items measuring the same construct, they could not be meaningfully calculated for this instrument. Reliability and validity were therefore not formally tested and are addressed as limitations below.

The following research questions guided the analysis:

1. What levels of knowledge and personal experience do Austrian secondary school students report regarding cybercrime and associated digital threats?
2. How well do students understand and practice digital self-protection, and to what extent do they feel prepared to deal with cybercrime and online risks?
3. What are students' perspectives on the current and future role of schools in providing education on cybercrime and internet safety?

Data analysis was conducted descriptively, presenting absolute and relative frequencies of responses for all closed-ended items. In addition, chi-square tests of independence were applied to examine whether key variables (e.g., awareness of cybercrime, personal experience, self-protection knowledge, or preparedness) differed significantly by gender and age. Cramér's *V* was calculated to estimate effect sizes.

For statistical comparisons by gender, only students identifying as male or female were included, as the groups identifying as diverse or with no information were too small for reliable testing. Nonetheless, their descriptive results were retained and are considered in the interpretation. To examine differences in preparedness more closely, independent samples t tests and Mann–Whitney U tests were additionally conducted. All quantitative analyses were performed using SPSS Statistics version 30. Open-ended responses were analyzed using qualitative content categorization, following Braun and Clarke's [58] thematic analysis framework, to identify central themes and perspectives expressed by the participants.

3.2 Survey implementation and ethical considerations

The data collection was conducted between February and the end of April 2025. Prior to administering the survey, ethical approval was obtained from the regional education authorities (*Bildungsdirektionen*) of Burgenland, Carinthia (*Kärnten*), Styria (*Steiermark*), Upper Austria (*Oberösterreich*), Tyrol (*Tirol*), and Vorarlberg. The study was conducted in compliance with national standards for educational research and data protection. Participation was strictly voluntary and anonymous. Students were clearly informed that their responses would not be linked to their identities or schools.

Due to administrative constraints and limited resources, the federal states of Vienna, Salzburg, and Lower Austria were not included in the study. After securing the necessary approvals, school administrations in the participating regions were contacted and provided with a link to the online questionnaire. The schools then distributed the survey to students aged 16 years and older.

3.3 Sample description

A total of $N = 843$ students participated in the study. Of these, $n = 656$ students (77.8%) completed the questionnaire in full, whereas $n = 187$ students (22.2%) submitted partially completed responses. For the purposes of analysis, only fully completed questionnaires were included to ensure a consistent and reliable dataset.

3.4 Demographic composition of the sample

The demographic profile of the sample was based on participants' age, type of school attended, and gender. In terms of age distribution, the sample was heavily concentrated in the 16- to 18-year-old range, with 16-year-olds forming the largest group ($n = 209$, 31.9%), followed by 18-year-olds ($n = 188$, 28.7%) and 17-year-olds ($n = 156$, 23.8%). Older students were less represented: 19-year-olds ($n = 66$, 10.1%), 20-year-olds ($n = 17$, 2.6%), and those older than 20 years ($n = 20$, 3.1%).

The type of school attended also varied, though the majority of respondents were enrolled in vocational education institutions. Specifically, $n = 404$ students (61.6%) attended a vocational middle or higher secondary school (German: BMHS), whereas $n = 209$ students (31.9%) were in vocational schools (German: Berufsschule). A smaller portion of the sample attended an academic secondary school ($n = 41$, 6.3%; German: Gymnasium), and only two participants (0.3%) reported attending a polytechnic school.

Regarding gender distribution, there was a slight overrepresentation of male participants: $n = 396$ students (60.3%) identified as male, $n = 243$ students (37.0%) identified

as female, $n = 6$ students (0.9%) selected a diverse gender option, and $n = 11$ students (1.7%) chose not to provide gender information.

4 Results

4.1 Awareness and personal exposure to cybercrime

Of the 656 Austrian secondary school students surveyed, the vast majority demonstrated a high level of general awareness of cybercrime. Specifically, 586 students (89.3%) reported that they knew what the term *cybercrime* means, while 40 (6.1%) were unsure and 30 (4.6%) indicated that they did not know the term.

In terms of personal exposure, 83 students (12.7%) stated that they had personally been affected by a cybercrime incident, and 216 (32.9%) reported that someone in their immediate environment had been affected. Taken together, 299 students (45.6%) had encountered cybercrime directly or indirectly, whereas 357 (54.4%) reported no experience of this kind.

When asked about their familiarity with specific digital threats, 610 respondents (93.0%) indicated that they had heard of fake news, while 598 (91.2%) were aware of manipulated videos or images, commonly referred to as deepfakes. Additionally, 416 students (63.4%) reported awareness of digital messages or content designed to spy on them. Only 14 participants (2.1%) indicated that they had not heard of any of the listed threats.

Among the 639 students identifying as female or male, the majority demonstrated a high level of awareness of the term *cybercrime* (see Table 1). Specifically, 91.9% of male students and 85.6% of female students reported knowing the term, whereas 9.9% of females and 3.5% of males indicated uncertainty, and 4.5% in both groups stated that they did not know the term. A chi-square test revealed that these differences were statistically significant, $\chi^2(2, N = 639) = 10.86, p = 0.004$, although the strength of the association was weak ($V = 0.13$).

Regarding personal exposure to cybercrime (see Fig. 1), 14.9% of male students reported having been directly affected compared to 7.4% of female students. Indirect exposure, defined as knowing someone who had been affected, was reported by 33.3% of males and 32.1% of females. A majority of females (60.5%) indicated no experience with cybercrime, compared to 51.8% of males. The chi-square test indicated that these differences were statistically significant, $\chi^2(2, N = 639) = 9.17, p = 0.010$, with a small effect size ($V = 0.12$).

Table 1 Awareness and personal experience with cybercrime by gender (female/male)

Variable	Female (n = 243)	Male (n = 396)	χ^2 (df, N)	p	Cramér's V
<i>Awareness of cybercrime</i>			10.86 (2, 639)	0.004	0.13
Knows term (%)	85.6	91.9			
Does not know (%)	4.5	4.5			
Unsure (%)	9.9	3.5			
<i>Personal experience with cybercrime</i>			9.17 (2, 639)	0.010	0.12
Direct victim (%)	7.4	14.9			
Indirect victim (%)	32.1	33.3			
No experience (%)	60.5	51.8			
<i>Direct or indirect experience</i>			4.64 (1, 639)	0.031	0.09
Yes (%)	39.5	48.2			
No (%)	60.5	51.8			

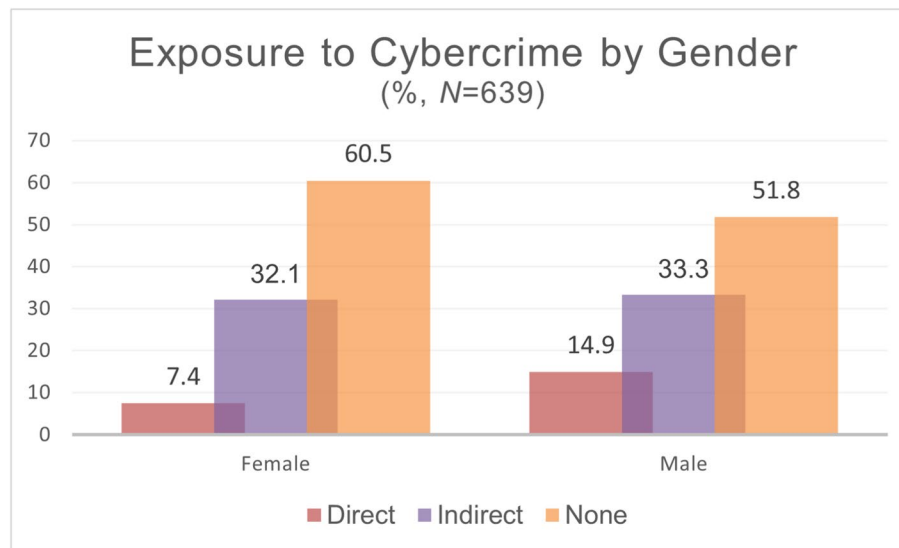


Fig. 1 Exposure to cybercrime by gender (%; N = 639)

Table 2 Cross-tabulation of age with knowledge and experience of cybercrime

Age	Knowledge of cybercrime (%)			Experience with cybercrime (%)		
	Yes	No	Not sure	Self-affected	Acquaintance	None
16 years (n = 209)	85.6	7.2	7.2	11.5	31.6	56.9
17 years (n = 156)	89.7	2.6	7.7	10.9	31.4	57.7
18 years (n = 188)	93.1	2.1	4.8	12.8	29.8	57.4
19 years (n = 66)	92.4	4.5	3.0	18.2	40.9	40.9
20 years (n = 17)	88.2	5.9	5.9	11.8	58.8	29.4
> 20 years (n = 20)	80.0	15.0	5.0	20.0	40.0	40.0
Total (N = 656)	89.3	4.6	6.1	12.7	32.9	54.4

Knowledge: Pearson's χ^2 (10, N = 656) = 15.28, p = 0.122, Cramér's V = 0.11

Experience: Pearson's χ^2 (10, N = 656) = 14.70, p = 0.144, Cramér's V = 0.11. Linear-by-linear association χ^2 (1, N = 656) = 6.27, p = 0.012

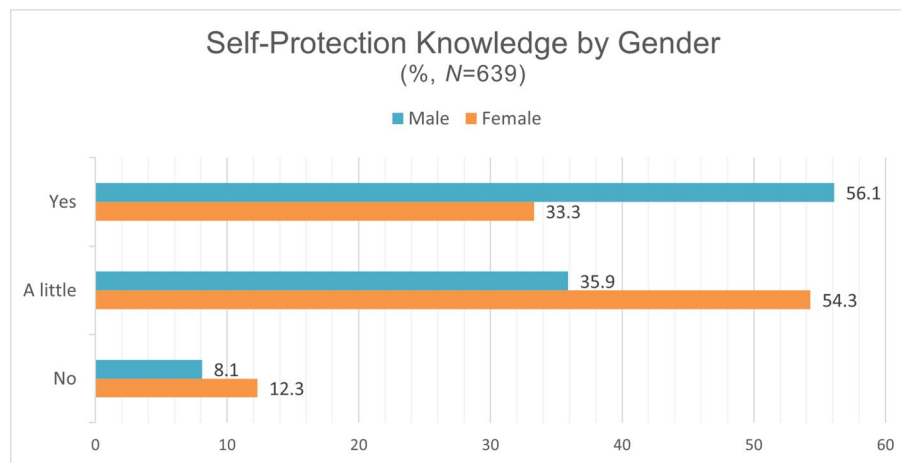
When direct and indirect experiences were combined, 48.2% of male students reported some form of exposure compared to 39.5% of female students. Conversely, 60.5% of females and 51.8% of males reported no exposure. This difference was statistically significant, $\chi^2(1, N = 639) = 4.64, p = 0.031$, though the effect size was very small ($V = 0.09$, equivalent to Φ for a 2×2 table).

When analyzed by age, no significant differences were found in students' knowledge of cybercrime (see Table 2). Awareness levels were consistently high across all groups, ranging from 85.6% among 16-year-olds to 93.1% among 18-year-olds, $\chi^2(10, N = 656) = 15.28, p = 0.122, V = 0.11$. This indicates that students, regardless of age, demonstrated a similarly strong understanding of the concept.

In terms of personal or indirect experience with cybercrime (see Table 2), no significant association with age was detected, $\chi^2(10, N = 656) = 14.70, p = 0.144, V = 0.11$. The Linear-by-Linear Association indicated a slight trend, $\chi^2(1, N = 656) = 6.27, p = 0.012$, with direct victimization somewhat more common among students aged 19 and older (18–20%) compared to younger students (approximately 11–13%). Given the small number of participants in the oldest groups, these results should be interpreted with caution.

Table 3 Self-protection knowledge ('Do you know how to protect yourself from cybercrime?') and preparedness ('How well do you feel prepared for digital dangers?') by gender (female/male)

Variable	Female (n = 243)	Male (n = 396)	χ^2 (df, N)	p	Cramér's V
Knowledge of self-protection			31.20 (2, 639)	< 0.001	0.221
No (%)	12.3	8.1			
A little (%)	54.3	35.9			
Yes (%)	33.3	56.1			
Preparedness for digital threats			11.96 (3, 639)	0.008	0.14
Not at all (%)	2.5	4.0			
Rather poorly (%)	28.0	18.2			
Rather well (%)	54.3	55.1			
Very well (%)	15.2	22.7			

**Fig. 2** Knowledge of self-protection by gender (N = 639)

4.2 Self-protection knowledge and preparedness

Understanding how students perceive their ability to protect themselves from cybercrime is a key step in assessing digital resilience. The survey data reveal important differences by gender and age, particularly regarding knowledge of protective strategies and self-assessed preparedness.

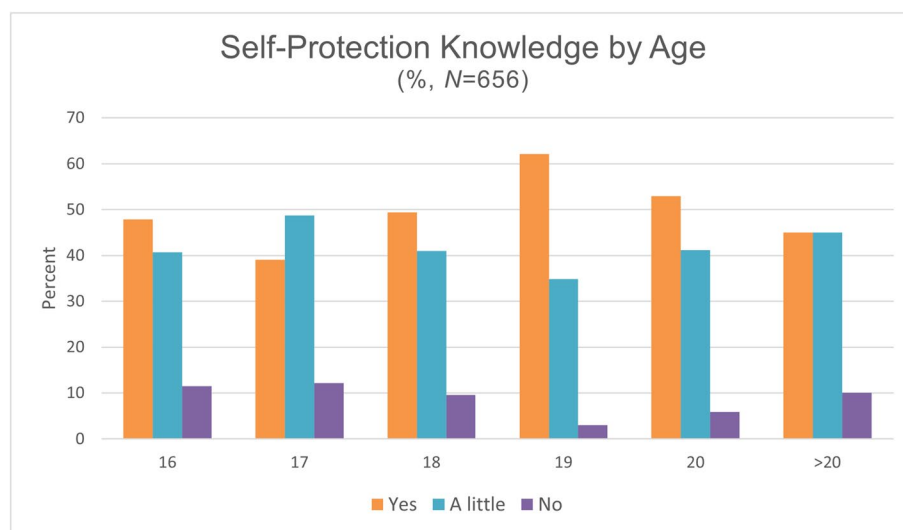
When asked whether they knew how to protect themselves from cybercrime, 313 of the 656 students (47.7%) reported that they did, while 277 (42.2%) indicated that they knew “a little.” Only 66 students (10.1%) indicated that they had no idea how to protect themselves.

In regard to self-protection, clear gender differences emerged (see Table 3). Of all male students, 56.1% reported knowing how to protect themselves from cybercrime, compared to only 33.3% of female students. Conversely, 54.3% of female students indicated knowing “a little,” compared to 35.9% of male students. Among students who indicated they did not know how to protect themselves, the proportion was somewhat higher among female students (12.3%) than male students (8.1%). The chi-square test indicated a highly significant association between gender and self-protection knowledge, $\chi^2(2, N = 639) = 31.20$, $p < 0.001$, with a moderate effect size ($V = 0.221$). This suggests that male students were more likely to feel confident in their protective knowledge, while female students were less likely to state they knew how to protect themselves (see Fig. 2).

Table 4 Cross-tabulation of age with self-protection knowledge and preparedness

Age group	Knowledge of self-protection (%)			Preparedness for digital threats (%)			
	Yes	A little	No	Not at all	Rather poorly	Rather well	Very well
16 y (n = 209)	47.8	40.7	11.5	4.3	20.1	57.4	18.2
17 y (n = 156)	39.1	48.7	12.2	5.1	23.7	54.5	16.7
18 y (n = 188)	49.5	41.0	9.6	0.5	23.4	53.2	22.9
19 y (n = 66)	62.1	34.8	3.0	4.5	19.7	53.0	22.7
20 y (n = 17)	52.9	41.2	5.9	5.9	11.8	47.1	35.3
> 20 y (n = 20)	45.0	45.0	10.0	20.0	15.0	45.0	20.0
Total (N = 656)	47.7	42.2	10.1	4.0	21.5	54.4	20.1

Knowledge of self-protection: Pearson's $\chi^2(10, N = 656) = 12.88, p = 0.230$, Cramér's $V = 0.099$. Preparedness: Pearson's $\chi^2(15, N = 656) = 26.59, p = 0.032$, Cramér's $V = 0.116$

**Fig. 3** Knowledge of self-protection by age (%; N = 656)

When comparing knowledge across age groups, the differences were not statistically significant (see Table 4). While descriptively, 16- and 18-year-olds reported higher knowledge of how to protect themselves, and 17-year-olds somewhat lower, the overall chi-square test was not significant, $\chi^2(10, N = 656) = 12.88, p = 0.230$, with a very small effect size ($V = 0.099$). Thus, knowledge of protective strategies appears relatively stable across age groups (see Fig. 3).

Students also reported engaging in concrete protective practices. The majority indicated they adopt at least some basic strategies (see Fig. 4):

- 564 students (86.0%) said they use secure passwords.
- 535 students (81.6%) reported avoiding links from strangers.
- 385 students (58.7%) stated that they regularly update their apps and programs.
- By contrast, only 150 students (22.9%) mentioned turning to parents or teachers when uncertain.

This low reliance on adult guidance suggests that students primarily depend on their own strategies or peer knowledge. In addition, 48 students (7.3%) listed “other” strategies, including VPNs, ad blockers, password managers, antivirus software, anonymous browsers, and careful data sharing. Some responses reflected sophisticated technical

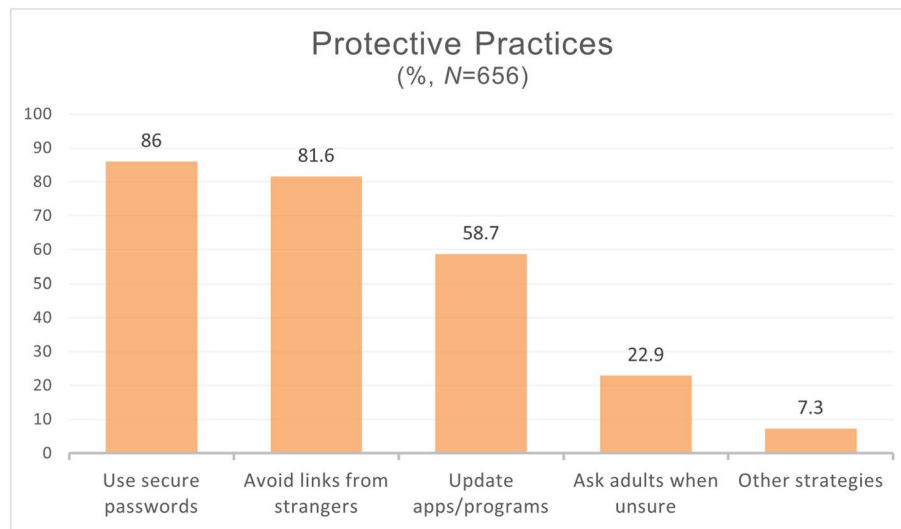


Fig. 4 Protective practices (% N = 656)

practices, while others were humorous or dismissive, showing varying levels of digital literacy and seriousness.

When asked how well they felt prepared to deal with digital dangers, the responses were varied. Most students felt at least somewhat prepared: 357 students (54.4%) reported feeling “rather well” prepared, and 132 (20.1%) “very well” prepared. On the other hand, 141 students (21.5%) felt “rather poorly” prepared, and 26 students (4.0%) said they were “not at all” prepared.

A significant gender difference was also observed here (see Table 3). Of all male students, 22.7% reported feeling very well prepared, compared to 15.2% of female students. A similar proportion of both groups reported feeling rather well prepared (55.1% of male and 54.3% of female students). Conversely, female students were more likely to report feeling rather poorly prepared (28.0%) than male students (18.2%), and a slightly higher proportion of male students (4.0%) than female students (2.5%) indicated feeling not at all prepared. The chi-square test confirmed a significant association between gender and perceived preparedness, $\chi^2(3, N = 639) = 11.96, p = 0.008$, with a small effect size ($V = 0.14$). While the majority of both male and female students felt at least rather well prepared, male students were more likely to report feeling very well prepared, whereas female students were disproportionately represented among those feeling rather poorly prepared (see Fig. 5).

Perceived preparedness varied somewhat across age groups (see Table 4). Descriptively, the proportion of students feeling very well prepared was lowest among 16- and 17-year-olds (18.2% and 16.7%, respectively), and somewhat higher among 18- and 19-year-olds (22.9% and 22.7%). The share of students reporting poor preparedness was relatively stable across the younger age groups, ranging from 20.1% among 16-year-olds to 23.7% among 17-year-olds and 23.4% among 18-year-olds, and declined somewhat among older students (19.7% for 19-year-olds). Statistical testing showed a small but significant association, $\chi^2(15, N = 656) = 26.59, p = 0.032$, with a small effect size ($V = 0.116$). Thus, perceived preparedness shows a modest tendency to increase with age, though the pattern is not strictly linear and should be interpreted with caution given the small sample sizes in older age groups.

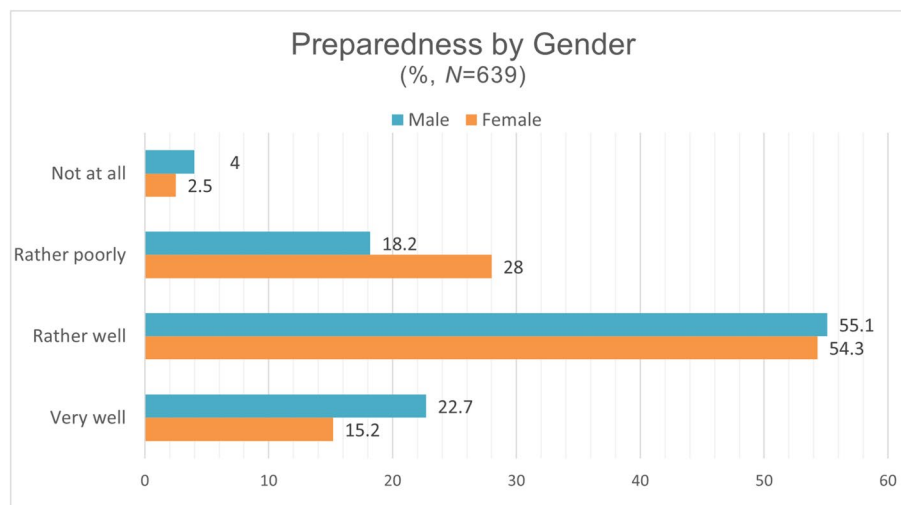


Fig. 5 Preparedness for digital threats by gender (N = 639)

4.3 Perceptions of school education on cybercrime and desires for improvement

For the third research question, students were asked about the role of schools in addressing cybercrime and internet safety. Despite a generally high awareness of cybercrime, only 391 students (59.6%) reported having received education on this topic at school, while 265 students (40.4%) stated that they had not. At the same time, 472 students (72.0%) expressed a clear desire to learn more about how to protect themselves online, whereas 184 students (28.0%) indicated that the topic did not interest them.

Among the 639 students identifying as female or male, analyses revealed no significant gender differences in whether students had received information about cybercrime at school ($\chi^2(1, N = 639) = 0.01, p = 0.935, \Phi = 0.003$), with 59.7% of female and 59.3% of male students reporting having received such education. Similarly, no significant gender difference emerged regarding interest in learning more ($\chi^2(1, N = 639) = 1.07, p = 0.301, \Phi = 0.041$), with 74.5% of female and 70.7% of male students expressing such interest. Furthermore, age was not significantly related to having received school-based education, $\chi^2(5, N = 656) = 4.15, p = 0.529, V = 0.079$. However, there was a small but significant age effect regarding students' desire for more education, $\chi^2(5, N = 656) = 11.19, p = 0.048, V = 0.131$, with interest remaining high among students aged 16 to 20 years (ranging from 66.5 to 77.3%), and declining notably only among students older than 20 years (50.0%), though this group was very small ($n = 20$) and results should be interpreted with caution (see Table 5).

A more pronounced difference was found when comparing preparedness between students who had or had not received school-based education. Students who reported receiving information in school felt significantly better prepared for digital dangers ($M = 1.93, SD = 0.65$) than those who had not ($M = 2.33, SD = 0.83$, on a scale from 1 = *very well prepared* to 4 = *not at all prepared*), $t(475.15) = -6.62, p < 0.001$, Cohen's $d = 0.55$. This result was confirmed by a Mann–Whitney U test, $U = 37,828.50, Z = -6.48, p < 0.001$, underscoring the robust association between school-based education and higher levels of perceived preparedness (see Table 6).

To further explore students' expectations, the questionnaire included an open-ended question on how schools could improve their education on artificial intelligence and internet safety. 655 out of 656 students provided a written response to this question. The

Table 5 cross-tabulation of age with school-based cybercrime education and desire for more information

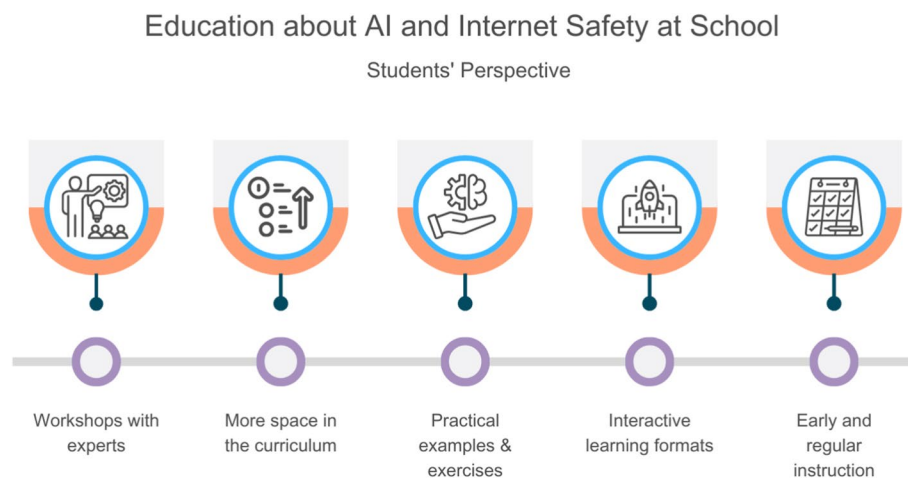
Age group	Cybercrime education at school (%)		Desire for more information (%)	
	Yes	No	Yes	No
16 y (n = 209)	56.0	44.0	66.5	33.5
17 y (n = 156)	60.3	39.7	73.7	26.3
18 y (n = 188)	61.2	38.8	76.6	23.4
19 y (n = 66)	68.2	31.8	77.3	22.7
20 y (n = 17)	58.8	41.2	76.5	23.5
> 20 y (n = 20)	50.0	50.0	50.0	50.0
Total (N = 656)	59.6	40.4	72.0	28.0

Cybercrime education: Pearson's $\chi^2(5, N = 656) = 4.15, p = 0.529$, Cramér's $V = 0.079$. Desire for more information: Pearson's $\chi^2(5, N = 656) = 11.19, p = 0.048$, Cramér's $V = 0.131$

Table 6 Perceived preparedness for digital dangers among students with versus without school-based cybercrime education

	Yes (n = 391)	No (n = 265)	Test/effect
Preparedness (M, SD)	1.93 (0.65)	2.33 (0.83)	$t(475.15) = -6.62, p < 0.001, d = 0.55$
Ranks (mean)	364.25	275.75	$U = 37,828.5, Z = -6.48, p < 0.001$

Preparedness was coded as 1 = *very well prepared* to 4 = *not at all prepared*; lower scores indicate higher perceived preparedness. Both parametric and nonparametric tests indicate higher preparedness among students who received information

**Fig. 6** Summary of students' expectations for school-based education on digital safety

answers revealed a strong and widespread demand for more interactive, relevant, and hands-on forms of education. Frequently mentioned suggestions included regular workshops, ideally involving external experts, greater integration of these topics into existing curricula, and even the introduction of dedicated subjects. Students repeatedly called for the use of real-life case studies, interactive exercises, and updated teaching methods that reflect their digital everyday lives. Many expressed a desire for more honest, practical guidance rather than generic warnings or outdated presentations. A small portion of responses also conveyed frustration, scepticism, or irony, but these often underscored the students' deeper wish to be taken seriously and to receive meaningful education about the digital world (see Fig. 6).

5 Discussion

The results of this study provide a differentiated view of Austrian secondary school students' awareness, experiences, and educational needs in relation to cybercrime. The overall high level of awareness, with almost 90% of students stating they know the term *cybercrime*, suggests that the concept is widely present in young people's everyday media environment. However, this finding is based on self-reported familiarity and should therefore be interpreted with caution, as students may overestimate their understanding or interpret the term differently depending on prior exposure through media or school contexts. This surface-level familiarity must be critically reflected upon. Knowing a term does not mean being fully informed about its real-life implications, especially in emotionally complex or socially embedded situations. Empirical studies assessing cybercrime awareness among young people report similarly high levels of general familiarity with cybercrime terminology. Studies conducted among youths in India and Nigeria likewise found that the vast majority of respondents reported awareness of cybercrime concepts or common digital threats [31, 32].

A large proportion of respondents, nearly half, reported having been personally affected by cybercrime or knowing someone who has. This confirms that cybercrime is a reality in the everyday lives of young people. It underlines the relevance of focusing on adolescents as a particularly vulnerable group, as emphasized by previous research [2, 7, 9]. These findings support the need for early, consistent, and targeted prevention strategies. Comparable youth research has similarly identified cyberbullying, hacking, and online blackmail as frequently experienced forms of cybercrime exposure among young people [35]. Large-scale European survey data from EU Kids Online further demonstrate that substantial proportions of adolescents encounter online risks even though not all exposures result in perceived harm [27]. The present findings therefore align with broader international evidence indicating that exposure to digital risks forms a regular component of adolescents' online environments.

Importantly, in this study, age differences in direct or indirect victimization were minor, with only a slight trend toward higher victimization among older students. This suggests that vulnerability is not confined to a particular age group, although developmental differences may still influence how students cope with risks.

Gender-specific differences in awareness, victimization, and self-protection emerged as an important dimension of the results. Male students reported higher awareness of the term *cybercrime*, greater direct victimization, and higher self-reported ability to protect themselves. Female students, by contrast, were more likely to describe their preparedness as *rather poor* and to indicate that they knew only *a little* about protecting themselves online. These findings align with international research showing that male adolescents often report greater risk-taking behavior, whereas female adolescents display higher perceived vulnerability and worry about online harms [59]. They also reflect broader patterns in digital self-efficacy, where female students tend to underestimate their competence despite equal or better performance on objective measures [60–62].

European survey evidence provides additional context for interpreting these differences. Findings from EU Kids Online indicate that gender differences in online risk exposure are often linked to patterns of online activities rather than overall internet use. Boys more frequently report contact with unknown persons and certain data misuse risks associated with gaming and interactive online environments, while girls tend to

report greater sensitivity to interpersonal online harms. Such activity-related differences may partly explain the higher exposure reported by male students alongside lower perceived preparedness among female students in the present study [27].

The high level of student familiarity with specific threats, such as fake news (93%) and deepfakes (91%), shows that young people are aware of the changing digital environment and emerging dangers, including those connected to generative AI [50, 52]. These findings suggest a broadly favorable starting point for targeted digital literacy education. However, awareness of threats is only one dimension of digital competence. When asked about their actual ability to protect themselves, only 47.7% felt they had sufficient knowledge, and a further 42.2% felt they knew *a little*. This indicates that there is still considerable uncertainty—particularly among female students, whose lower self-assessed competence mirrors international findings that women and girls tend to report lower cybersecurity self-efficacy [63].

Recent European cybersecurity education research similarly highlights a discrepancy between students' perceived confidence and their actual cybersecurity practices. Findings from the CONCORDIA survey indicate that many high-school students across EU countries report feeling confident in detecting or handling online risks, yet frequently fail to select correct security practices when confronted with concrete cybersecurity scenarios. These results suggest that perceived competence may not necessarily translate into effective protective behaviour and underline the importance of structured school-based cybersecurity education [28].

At the behavioral level, students mostly engage in basic digital safety practices. For example, 86.0% use strong passwords, and 81.6% avoid clicking on suspicious links. These habits suggest that fundamental technical knowledge is widespread. However, seeking help from adults is less common: only about 23.0% of students said they turn to teachers or parents when unsure. This reflects a critical weakness in help-seeking behavior and social support systems, especially when facing emotionally charged forms of cybercrime such as sextortion, cyber grooming, or doxing. Comparable patterns are visible in European youth research. The EU Kids Online study reports that a proportion of children do not disclose negative online experiences to anyone, with rates ranging from 4 to 30% across countries. When disclosure occurred, children most often turned to parents or friends, while teachers or professional support services were mentioned only rarely [27]. This suggests that schools and formal support systems may currently play a more limited role in adolescents' coping strategies compared to informal support networks. European findings from the CONCORDIA survey similarly indicate that many students do not share online risk experiences with parents or teachers, suggesting limited reliance on formal support structures when dealing with cybersecurity incidents. Strengthening collaboration between students, educators, and families was therefore identified as an important objective of school-based cybersecurity education initiatives [28]. Concerns about embarrassment, fear of losing digital autonomy, or uncertainty regarding adult reactions may contribute to this pattern. These forms of attack are often interpersonal and deeply psychological, meaning technical solutions alone are not enough. Here, the concept of *digital resilience* becomes crucial—the ability not only to avoid risks, but to respond, recover, and seek support [42, 48].

Although most students reported feeling at least somewhat prepared to deal with digital dangers (74.5%, combining those who felt rather well and very well prepared), the

gender differences in perceived preparedness were significant. Male students more often felt *very well prepared*, whereas female students more frequently rated themselves as poorly prepared. Age differences were smaller, though older students tended to report higher preparedness. These findings show that confidence in handling digital threats is unevenly distributed and suggest that empowerment efforts should pay close attention to gender-related gaps in confidence and perceived competence.

A key finding of this study is the substantial gap between students' needs and what schools currently provide. While 59.6% of students had received education on cybercrime at school, 72.0% expressed a desire to learn more. Interest in further education remained high across most age groups, declining notably only among students older than 20 years, though this group was very small and results should be interpreted with caution. Importantly, those who had received school-based education felt significantly better prepared to deal with digital dangers, with a medium effect size. This highlights the positive impact of structured school-based interventions on students' digital resilience.

This finding corresponds with international evidence emphasizing the importance of educational contexts for developing digital competence and risk awareness. Large-scale comparative assessments such as ICILS identify substantial disparities in students' digital skills and underline the role of education systems in addressing these inequalities [30]. European cybersecurity education initiatives further emphasize the central role of teachers and structured curricula in promoting risk awareness and coping strategies. Evidence from the CONCORDIA project highlights the need for coordinated and interactive cybersecurity education approaches that actively involve students, teachers, and parents in addressing online risks [28].

The open-ended responses reinforce this demand: nearly all students offered suggestions for improving education, calling for more interactive, relevant, and hands-on learning formats. Workshops, expert contributions, real-life case studies, and integration into everyday subjects were frequently emphasized. These demands align with national curricula such as *Digitale Grundbildung* and international frameworks that promote participatory, meaningful, and prevention-oriented digital education [41, 46]. Designing such programs in close alignment with students' lived experiences can ensure that they are both relevant and effective.

Overall, the findings suggest that Austrian students are highly aware of cybercrime and already apply some protective strategies. However, significant gender differences in self-assessed knowledge and preparedness remain, and schools currently do not meet students' expressed demand for meaningful education on digital risks. Strengthening digital resilience requires not only technical literacy but also emotional, social, and ethical dimensions of education. A gender-inclusive approach that addresses specific vulnerabilities and confidence gaps will be essential in ensuring that all students are empowered to navigate the digital world safely.

6 Limitations

This study also has several limitations. First, due to administrative restrictions, only six Austrian federal states were included, which may affect the generalizability of the findings. Second, the results are based on self-assessments and voluntary responses, which can lead to bias or inaccuracies. In particular, students' responses regarding whether

they know the term *cybercrime* reflect self-reported familiarity, as no additional objective measures or knowledge tests were conducted to verify their level of understanding. Consequently, the reported awareness should be interpreted cautiously and cannot be equated with demonstrated conceptual knowledge or competencies. This interpretation is supported by patterns observed in the qualitative data, as some open-ended answers suggest disengagement, irony, or limited understanding. Third, the questionnaire was purpose-built for this study and pretested internally with researchers and university student staff members as age-proximate reviewers, rather than with secondary school students from the target population. Neither formal reliability testing (not applicable given the single-item structure of most measures) nor external psychometric validation was conducted. This limits conclusions about the instrument's precision and generalizability and should be addressed through a validated scale in future research. Finally, the study measures perceptions and attitudes but not actual competencies. Future research should use mixed methods, including interviews, classroom observations, and objective skill assessments, to better capture practical abilities, emotional coping strategies, and how these differ across male, female, diverse, and non-binary students.

7 Conclusion

This study explored Austrian secondary school students' perspectives on cybercrime and digital self-protection. By examining their knowledge, personal experiences, protective behaviors, and expectations for school-based education, the study offers a broad understanding of how adolescents navigate digital risks and what support they need in doing so.

The findings show that while many students are familiar with the concept of cybercrime and already apply basic protective strategies, gaps remain when it comes to deeper knowledge, emotional coping, and structured prevention. Notably, the results revealed clear gender differences: male students reported higher awareness, greater self-assessed preparedness, and more confidence in protective abilities, whereas female students more often indicated uncertainty or limited preparedness. Age-related patterns were weaker but suggested that older students tend to feel slightly more prepared, while interest in further education remained consistently high across most age groups. Students identifying as diverse or giving no information regarding gender, though fewer in number, represent an important group with potentially distinct risk profiles and support needs. A significant number of students across all gender identities reported direct or indirect experiences with cybercrime, confirming that digital threats are part of their everyday lives.

At the same time, students express a clear and consistent wish for more education on this topic, particularly in the form of practical, interactive, and up-to-date learning opportunities. Importantly, those who had already received school-based education felt significantly better prepared to deal with digital dangers, highlighting the positive impact of structured interventions. These results underline the central role of schools in preparing young people for a connected and complex digital world. In line with existing educational frameworks, including Austria's *Digitale Grundbildung*, cybercrime education must move beyond technical instructions to include critical thinking, social-emotional learning, and digital resilience. Such approaches should be gender-inclusive, addressing

the specific risks, confidence gaps, and help-seeking behaviors of male, female, diverse, and non-binary students.

Overall, this study highlights both the challenges and opportunities for digital education in schools. It provides valuable insight into students' lived realities and contributes to the ongoing discussion about how to strengthen digital literacy and safety in education. Developing well-planned and inclusive programmes that reflect students' actual concerns and circumstances allows schools to use their resources efficiently, while also fulfilling their responsible role in preparing all young people for a complex digital world. By listening to young people's voices and addressing their needs, schools can play a key role in helping them develop the skills and confidence to navigate digital spaces safely, responsibly, and with resilience.

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Author contributions

M.U. Contributor Roles: Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Writing—original draft, Writing—review and editing, Resources, Visualization, Formal analysis, Validation C.H. Contributor Roles: Conceptualization, Funding acquisition, Investigation, Methodology, Writing—original draft, Writing—review and editing, Resources, Formal analysis, Validation L.K. Contributor Roles: Conceptualization, Writing—original draft, Writing—review and editing, Validation B.S. Contributor Roles: Supervision, Project administration, Funding acquisition, Validation.

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Data availability

The author confirms that all data related to cybercrime generated or analysed during this study are fully included in this published article. The data were collected through a survey comprising two thematic sections: (1) Artificial Intelligence in Schools and (2) Cybercrime and Internet Safety. In line with the specific focus of this study on cybercrime in educational contexts, only data from the second section were analysed and are presented here. Data from the first section, concerning artificial intelligence in schools, are reported separately in a related publication (<https://doi.org/10.1016/j.ssaho.2026.102821>).

Declarations

Ethics approval and consent to participate

Data collection was conducted between February and the end of April 2025. Prior to administering the survey, ethical approval was obtained from the regional education authorities (*Bildungsdirektionen*) of Burgenland, Carinthia (*Kärnten*), Styria (*Steiermark*), Upper Austria (*Oberösterreich*), Tyrol (*Tirol*), and Vorarlberg. The study was conducted in compliance with national standards for educational research and applicable data protection regulations. Participation in the study was strictly voluntary and anonymous. Students were clearly informed about the purpose of the study and that their responses would not be linked to their identities or schools. Informed consent to participate in the study was obtained from all participants prior to participation.

Consent for publication

Not applicable. The manuscript does not contain any individual person's identifiable data.

Transparency statement

During the development of this manuscript, various AI-assisted technologies were employed to support and improve the writing process, while maintaining strict adherence to academic standards. ChatGPT was utilized exclusively for language polishing, including correcting grammar and enhancing clarity. Additionally, Elicit was used as a resource to aid in literature review and identifying pertinent studies. Importantly, these AI tools did not contribute to generating original content, performing data analysis, or shaping the core arguments. The authors affirm full responsibility for the manuscript's authenticity and scholarly integrity.

Competing interests

The authors declare no competing interests.

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References

1. Aiken M, Davidson J, Amann P. Youth pathways into cybercrime. Europol White Paper; 2016. <https://www.europol.europa.eu/cms/sites/default/files/documents/pathways-white-paper.pdf>.
2. Feinberg T, Robey N. Cyberbullying. *Educ Dig*. 2009;74(7):26. <https://doi.org/10.1542/peds.2016-1758U>.
3. Douglas DM. Doxing: a conceptual analysis. *Ethics Inf Technol*. 2016;18(3):199–210. <https://doi.org/10.1007/s10676-016-9406-0>.
4. Moncur W, Orzech KM, Neville FG. Fraping, social norms and online representations of self. *Comput Hum Behav*. 2016;63:125–31. <https://doi.org/10.1016/j.chb.2016.05.042>.
5. Mathiesen A. Cybermobbing und Cybergrooming: Neue Kriminalitätsphänomene Im Zeitalter Moderner Medien. Kriminalwissenschaftliches Institut der Leibniz Universität Hannover, Hannover; 2014. <https://doi.org/10.15488/3677>.
6. Döring N. Erotischer fotoaustausch unter jugendlichen: Verbreitung, funktionen und folgen des sexting. *Zeitschrift für Sexualforschung*. 2012;25(1):4–25. <https://doi.org/10.1055/s-0031-128394>.
7. Wolak J, et al. Sextortion of minors: characteristics and dynamics. *J Adolesc Health*. 2018;62(1):72–9. <https://doi.org/10.1016/j.jadohealth.2017.08.014>.
8. Jakobsson M, Myers S, editors. Phishing and countermeasures: understanding the increasing problem of electronic identity theft. Hoboken: Wiley; 2007.
9. Coluccia A, et al. Online romance scams: relational dynamics and psychological characteristics of the victims and scammers: a scoping review. *Clin Pract Epidemiol Ment Health*. 2020;16:24. <https://doi.org/10.2174/1745017902016010024>.
10. Bundeskriminalamt B. Cybercrime Report 2023. Lagebericht über die Entwicklung von Cybercrime. Bundesministerium für Inneres. 2024. Accessed: 10 July 2025 https://www.bundeskriminalamt.at/306/files/Cybercrime_Report_2023_WebBF.pdf.
11. WHO/Europe: One in six school-aged children experiences cyberbullying, finds new WHO/Europe study. 2024. Accessed: 10 July 2025. <https://www.who.int/europe/news/item/27-03-2024-one-in-six-school-aged-children-experiences-cyberbullying-finds-new-who-europe-study>.
12. Süss D et al. Ergebnisbericht zur JAMES-Studie 2024; 2024. Accessed: 10 July 2025 https://www.swiss-schools.ch/wp-content/uploads/2024/11/JAMES-Studie_2024.pdf.
13. Beitzinger F, Leest U. Cyberlife V. Spannungsfeld zwischen Faszination und Gefahr - Cybermobbing bei Schülerinnen und Schülern. Bündnis gegen Cybermobbing e.V. 2024. Accessed: 10 July 2025. https://bueundnis-gegen-cybermobbing.de/wp-content/uploads/2024/10/Cyberlife_Studie_2024_Endversion.pdf.
14. Saferinternet.at: Studie: Cyber-Mobbing, Wien; 2022. Accessed: 10 July 2025 <https://www.saferinternet.at/presse-detail/studie-cyber-mobbing>.
15. Saferinternet.at: Jugend Internet Monitor 2025, Wien; 2025. Accessed: 10 July 2025 <https://www.saferinternet.at/services/jugend-internet-monitor>.
16. Huber E. Cybercrime: eine Einführung. Wiesbaden: Springer; 2019. <https://doi.org/10.1007/978-3-658-26150-4>.
17. Gordon S, Ford R. On the definition and classification of cybercrime. *J Comput Virol*. 2006;2:13–20. <https://doi.org/10.1007/s11416-006-0015-z>.
18. Commission E. Cybercrime. Online resource. Accessed: 10 July 2025. https://home-affairs.ec.europa.eu/policies/internal-security/cybercrime_en.
19. Bundeskriminalamt B. Delikte & Ermittlungen. Internetkriminalität. Online resource. Accessed: 10 July 2025. <https://www.bundeskriminalamt.at/306/start.aspx>.
20. Peebles E. Cyberbullying: hiding behind the screen. *Paediatr Child Health*. 2014;19(10):527–8. <https://doi.org/10.1093/pch/19.10.527>.
21. Peter IK, Petermann F. Cyberbullying: a concept analysis of defining attributes and additional influencing factors. *Comput Hum Behav*. 2018;86:350–66. <https://doi.org/10.1016/j.chb.2018.05.013>.
22. Murphy C. Cyberbullying among young people: laws and policies in selected member states. European Parliamentary Research Service; 2024. Accessed: 10 July 2025 [https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/762331/EPRS_BRI\(2024\)762331_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/762331/EPRS_BRI(2024)762331_EN.pdf).
23. Wachs S, Wolf KD, Pan CC. Cybergrooming. Risk factors, coping strategies and associations with cyberbullying. *Psicothema*. 2012;2012:628–33.
24. Ray A, Henry N. Sextortion: a scoping review. *Trauma Viol Abuse*. 2025;26(1):138–55. <https://doi.org/10.1177/15248380241277271>.
25. Nuccitelli M. 2012 cyber bullying tactics. *Forens Exam*. 2012;21(3):24.
26. Willard NE. Cyberbullying and cyberthreats: responding to the challenge of online social aggression, threats, and distress. Champaign: Research Press; 2007.
27. Smahel D, Machackova H, Mascheroni G, Dedkova L, Staksrud E, Ólafsson K, Livingstone S, Hasebrink U. EU Kids Online 2020: survey results from 19 countries.
28. Concordia: teaching cybersecurity in high school: survey report. https://www.concordia-h2020.eu/wp-content/uploads/2021/12/TEACHING-CYBERSECURITY-IN-HIGH-SCHOOL-survey_report.pdf.
29. Livingstone S, Helsper E. Balancing opportunities and risks in teenagers' use of the internet: the role of online skills and internet self-efficacy. *New Media Soc*. 2010;12(2):309–29. <https://doi.org/10.1177/146144480934269>.
30. European Commission. Directorate-General for Education, Youth, Sport and Culture: International Computer and Information Literacy Study (ICILS) in Europe, 2023: Main Findings and Educational Policy Implications. <https://doi.org/10.2766/5221263>.
31. Sani KM, Hassan MA, Danjuma AH, Adam MM, Nadama SG, Udu MS, et al. Study on cybercrime awareness among youths in Kaduna south local government area, Kaduna State, Nigeria. *Int J Indian Psychol*. 2024;12(4):1.
32. Meena Y, Sankhla M, Mohril S, Kumar R. Cybercrime: youth awareness survey in Delhi NCR, India. *Forensic Res Criminol Int J*. 2020;8(5):177–80. <https://doi.org/10.15406/frcij.2020.08.00325>.
33. Mankotia S, Sharma S. Awareness of young internet users towards cyber security and cyber crimes. *Int Res J Educ and Technol*. 2022;4(8):166–72.
34. Toso CHS, Jumalon AJA, Magadan JAR, Alvarico A Jr, Cuevas J. Cybercrime awareness among senior high school students. *Mediterranean J Basic Appl Sci (MJBAS)*. 2023;7(2):160–76. <https://doi.org/10.46382/MJBAS.2023.7218>.
35. Ullah F, Ali A, Umar Z. Understanding cybercrime and youth: a perception based approach. *Pak J Soc Res*. 2021;3(3):130–6.

36. Alsaeed HR, Elsayad WA, Abd El Ghani RT, Hassan MA. Awareness of cybercrime risks and its relationship to attitude toward the internet use among university students. *J Posit Sch Psychol*. 2023;7(10):1.
37. Maddux JE, Rogers RW. Protection motivation and self-efficacy: a revised theory of fear appeals and attitude change. *J Exp Soc Psychol*. 1983;19(5):469–79. [https://doi.org/10.1016/0022-1031\(83\)90023-9](https://doi.org/10.1016/0022-1031(83)90023-9).
38. Liang H, Xue Y. Avoidance of information technology threats: a theoretical perspective1. *MIS Q*. 2009;33(1):71–90. <https://doi.org/10.2307/20650279>.
39. Strafgesetzbuch [StGB]: Strafgesetzbuch in der geltenden Fassung. Online resource. Accessed: 10 July 2025. <https://www.ris.bka.gv.at/>.
40. OECD: Children in the digital environment: Revised typology of risks. Technical report 302, Paris: 2021. <https://doi.org/10.1787/9b8f222e-en>.
41. UNESCO: Technology in Education: A Tool on Whose Terms? Paris: UNESCO Publishing; 2023. <https://unesdoc.unesco.org/ark:/48223/pf0000380663>.
42. UNESCO: Global education monitoring report 2023: Technology in education—a tool on whose terms? Technical report. Paris: 2023. <https://unesdoc.unesco.org/ark:/48223/pf0000385723>.
43. Vuorikari, R., Kluzer, S., & Punie, Y. (2022). *DigComp 2.2, The Digital Competence framework for citizens : with new examples of knowledge, skills and attitudes*, Publications Office of the European Union. <https://data.europa.eu/doi/10.2760/115376>
44. N.N. Blinded for Review
45. Wirtschaftsstandort B-B. Digitales Kompetenzmodell für Österreich—DigComp 2.2 AT; 2021. https://www.fit4internet.at/media/digcomp2_2_pdf.
46. Bildung Wuf. Lehrplan für die Sekundarstufe I: Pflichtgegenstand Digitale Grundbildung (BGBl. II Nr. 267/2022); 2022. https://www.ris.bka.gv.at/Dokumente/BgblAuth/BGBLA_2022_II_267/BGBLA_2022_II_267.pdf.
47. OECD: Preparing teachers for digital education. Technical report. Paris; 2025. <https://doi.org/10.1787/af442d7a-en>.
48. Internet Safety UC. Digital Resilience Framework. GOV.UK. <https://www.gov.uk/government/publications/digital-resilience-framework>.
49. Stoilova M, Livingstone S, Khazbak R. Investigating risks and opportunities for children in a digital world: a rapid review of the evidence on children's internet use and outcomes. Innocenti discussion paper, UNICEF Office of Research—Innocenti; 2021. <https://ideas.repec.org/p/ucf/indipa/indipa1183.html>.
50. Neupane S, et al. Impacts and risk of generative ai technology on cyber defense; 2023. <https://doi.org/10.48550/arXiv.2306.13033>.
51. Economic E, Committee S. Generative AI and foundation models in the EU: uptake, opportunities, challenges, and a way forward. Technical report, Brussels: Centre for European Policy Studies; 2025.
52. Pater J, et al. A commentary on sexting, sextortion, and generative AI: risks, deception, and digital vulnerability. *Fam Relat*. 2025;74(3):1109–20. <https://doi.org/10.1111/fare.13152>.
53. Centre U, Foundation B. Generative AI: a new threat for online child sexual exploitation and abuse; 2024. <http://lab.unicri.it/Publication-Generative-AI-Threat-Child-Sexual-Exploitation-Abuse>.
54. Europol: 25 arrested in global hit against AI-generated child sexual abuse material; 2025. https://www.europol.europa.eu/media-press/newsroom/news/25-arrested-in-global-hit-against-ai-generated-child-sexual-abuse-material?mtm_campaign=newsletter.
55. Treleaven P, et al. The future of cybercrime AI and emerging technologies are creating a cybercrime tsunami; 2023. <https://doi.org/10.2139/ssrn.4507244>.
56. O'Connor K. AI-generated images: legal considerations of sexting and sextortion. *Fam Relat*. 2025;74(3):1109–20. <https://doi.org/10.1111/fare.13152>.
57. Redecker C. European framework for the digital competence of educators: Digcompedu. Technical report, Luxembourg: Publications Office of the European Union; 2017. <https://doi.org/10.2760/159770>.
58. Braun V, Clarke V. Using thematic analysis in psychology. *Qual Res Psychol*. 2006;3(2):77–101. <https://doi.org/10.1191/1478088706qp0630a>.
59. Trinidad A, Marcos V, Montes A, Seijo D. Negative online experiences, worry, and risk perception among adolescents: gender differences and implications for cybercrime awareness. *Cyberpsychol Behav Soc Netw*. 2025;28(5):348–53. <https://doi.org/10.1089/cyber.2024.0476>.
60. Mercader C, Duran-Bellonch M. Female higher education teachers use digital technologies more and better than they think. *Digit Educ Rev*. 2021;40:172. <https://doi.org/10.1344/der.2021.40.172-184>.
61. Siddiq F, Scherer R. Is there a gender gap? A meta-analysis of the gender differences in students' ICT literacy. *Educ Res Rev*. 2019;27:205–17. <https://doi.org/10.1016/j.edurev.2019.03.007>.
62. Aesaert K, Voogt J, Kuiper E, Braak J. Accuracy and bias of ICT self-efficacy: an empirical study into students' over-and underestimation of their ICT competences. *Comput Hum Behav*. 2017;75:92–102. <https://doi.org/10.1016/j.chb.2017.05.010>.
63. Anwar M, He W, Ash I, Yuan X, Li L, Xu L. Gender difference and employees' cybersecurity behaviors. *Comput Hum Behav*. 2017;69:437–43. <https://doi.org/10.1016/j.chb.2016.12.040>.
64. Unterweger M, Hörmann C, Kuka L, Sabitzer B. Empowering Young Women against cybercrime: building digital resilience in the age of AI manipulation. In: *EDULEARN25 proceedings. 17th international conference on education and new learning technologies*. Spain: IATED, Palma; 2025. p. 655–61. <https://doi.org/10.21125/edulearn.2025.0255>.

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